Six math clues · answer key

Shadows at Noon

Age 8 / Grade 2 stretch. Offer tens-and-ones drawings, repeated addition, or oral reasoning.

Math 1 · 6

3 + 3 = 6. Two rows of 3 make 6 chalk marks.

Math 2 · 12

4 + 4 + 4 = 12. Three groups of 4 make 12 measuring pebbles.

Math 3 · 8

2 + 2 + 2 + 2 = 8. Four rows of 2 make 8 sticks.

Math 4 · 10

5 + 5 = 10. Two rows of 5 make 10 notebook boxes.

Math 5 · 15

3 + 3 + 3 + 3 + 3 = 15. Five groups of 3 make 15 shadow-tip dots.

Math 6 · 25

5 + 5 + 5 + 5 + 5 = 25. Five rows of 5 make 25 chalk circles.

Age 8 / Grade 2 math. Six equal-group and array tasks use repeated addition. Rows and columns are each 2–5. Welcome drawings, spoken reasoning, or arranging small objects. Use the quantities in each question, rather than counting painted details.

Reading & writing · answer key

Shadows at Noon

Reading · 1

The sun was higher around noon, so the same standing stick cast a shorter shadow.

Reading · 2

The shadow points away from the light. The bright side of the stick faces the sun.

Reading · 3

Leave the stick standing in the same place. Mark each shadow tip, measure from the stick with the same unit, and compare the lengths.

Writing

Accept one sensible command, spoken, dictated, or written. Example: “Measure from the stick’s base to the shadow tip with the same unit.”

Age 8, Grade 3 reading, and Grade 2 math. This story uses a sunny yard at middle latitudes. Solar noon is when the sun reaches its highest point that day; it need not be 12:00 by the clock. For the same vertical stick on level ground, that gives the shortest daytime shadow. The sun’s height varies with place and season; in most of the United States the noon sun is never straight overhead. Clear sunlight gives a distinct edge; clouds can soften it. The diorama isolates sun height and light direction, with equal chalk spaces for comparison, not a clock or local forecast. The math counts equal groups and rows of garden and notebook objects. For an optional outdoor try, an adult helps place a stable blunt stick beside a path. Enjoy watching the shadow on the ground; never look at the sun.

Accept a spoken command, dictation, invented spelling, or writing. One clear sentence is enough. Example: “Measure from the stick’s base to the shadow tip with the same unit.”

Four shadow clues

Shadows at Noon

Quiz 1. In this sunny yard, what happens to the same stick’s shadow around noon?

Around noon, the sun is near its highest point for the day, so the same standing stick has a shorter shadow.

Quiz 2. What happens to the shadow when the sun is lower?

A lower sun makes a longer shadow from the same standing stick.

Quiz 3. Which way does a shadow point?

A shadow points away from the light. The bright side of the stick faces the sun.

Quiz 4. How can Little compare morning and noon shadows?

Keep the stick standing in the same place. Mark each tip and measure from the stick with the same unit, then compare the lengths.

Age 8, Grade 3 reading, and Grade 2 math. This story uses a sunny yard at middle latitudes. Solar noon is when the sun reaches its highest point that day; it need not be 12:00 by the clock. For the same vertical stick on level ground, that gives the shortest daytime shadow. The sun’s height varies with place and season; in most of the United States the noon sun is never straight overhead. Clear sunlight gives a distinct edge; clouds can soften it. The diorama isolates sun height and light direction, with equal chalk spaces for comparison, not a clock or local forecast. The math counts equal groups and rows of garden and notebook objects. For an optional outdoor try, an adult helps place a stable blunt stick beside a path. Enjoy watching the shadow on the ground; never look at the sun.